# README for Replication of "Lifestyle Behaviors and Wealth-Health Gaps in Germany"

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## 1. Overview

This document describes the corrected replication package for Mahler and Yum (2024), including the data used and the scripts that reproduce the corrected results reported in the corrigendum and the corresponding figures and tables in the paper and online appendix. This package supersedes the original replication package released with Mahler and Yum (2024). It incorporates the implementation corrections described in the corrigendum. The organization of the replication files and the instructions for running the programs are otherwise unchanged.

## 2. Data Availability and Provenance Statements

### Data Source

For this paper, we use confidential microdata from the German Socio-Economic Panel (SOEP). The raw data is from SOEP-Core, v35, doi:10.5684/soep-core.v35. These files are available to any researcher interested in replicating this analysis by following the detailed instructions in the "Data Availability Statement" section below.

### Statement about Rights

We certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

### Data Availability Statement

Data are available from the German Socio-economic Panel Study (SOEP) due to third party restrictions (for requests, please contact soepmail@diw.de). The scientific use file of the SOEP with anonymous microdata is made available free of charge to universities and research institutes for research and teaching purposes. The direct use of SOEP data is subject to the strict provisions of German data protection law. Therefore, signing a data distribution contract is a precondition for working with SOEP data. The data distribution contract can be requested with a form, available at: http://www.diw.de/soepforms. For further information, contact the SOEPhotline at either soepmail@diw.de or +49-30-89789-292.

## 3. Computational Requirements

The current replication package has been constructed and tested using Stata (version 19.5), Matlab (version 2024b), and Fortran (Intel Fortran Compiler (ifx) 2023.0 from Intel oneAPI with the Math Kernel Library, integrated in Microsoft Visual Studio 2022). The scripts were last tested on a Windows 11 PC with AMD Ryzen 9 7950X 16-Core Processor and 32 GB ram. The total running time for all scripts for the empirical parts using Stata takes less than one hour. The rest of the process can take several days. We provide estimates for the tasks that require substantial computing time below.

## 4. Instructions for Running the Scripts

The following instructions explain how to replicate all the figures and tables. We recommend that a replicator run the following instructions sequentially to ensure that preliminary information required for certain results is generated properly. As the replication requires us to use different softwares, the files are organized into different folders, which is clearly noted below.

- Download the replication folder. The replication folder contains seven subfolders:
    - dofiles: contains the Stata do-files for the empirical work we do using the SOEP data, producing some figures, tables, and auxiliary scripts used in the other programs
    - rawdata: this is where the raw data should be stored (see How to obtain the Data)

    - workdata: this is where the working data files are stored. Two files (workdata04-18.dta and survival_dat.dta) should be generated from confidential rawdata. The folder also contains five txt. files regarding model estimation, as detailed below.
    - fortran: contains the fortran files to solve and estimate the model
    - mfiles: contains the matlab scripts used to produce figures and tables
    - figures: the figures produced are saved in this folder
    - tables: the tables are saved in this folder
- Open the do-file "dofiles\master.do" and adjust the global "path" to where the replication folder is located. Using Stata, run "dofiles\master.do" to run all data work, produce figures A1 and A2 and export the data used in all tables in the paper and online appendix. Due to bootstrapping, this can take an hour or so.
- Using Matlab, open and run the "mfiles\prod_process.m" to estimate the autoregressive coefficient of the productivity process. This takes several seconds.
- Using Matlab, open and run the "mfiles\simul_mortality.m" file located in the mfiles folder to produce Table A10. This takes several seconds.
- This step is only required if you want to re-estimate the model starting from arbitrary initial guesses for the parameters. Using Visual Studio, open the "fortran\benchmark\Calibration.sln" file. For the point estimates of internally estimated parameters in Table 2, compile and solve after setting calind = 1 in the "fortran\benchmark\mod_parameters_v12.f90" file (line 10). On the machine we use, each evaluation takes less than two minutes. Depending on the initial parameter values, it could take 1,000 or more evaluations to obtain a satisfactory fit of the model, implying the total execution hours of 30 or more. The point estimates resulting from our estimation are stored in the "workdata\theta_init.txt" and the model-implied moments from the first estimation are stored in the "workdata\h_theta.txt". These files correspond to the values we have estimated (we have deemed satisfactory in terms of model fit) and may differ slightly from a re-estimation of the model.
- Using Matlab, open and run the "mfiles\get_standard_errors.m" to obtain the standard errors in Table 2. It requires the Jacobian matrix obtained from forward perturbation of each parameter, stored in "\workdata\G_fwd.txt" with a step length of 1e-3. These values will depend on the

estimation results from the previous step. For that reason, we have saved our estimation results in the txt files in the workdata folder. The two additional required files "mu_hat.txt" and "moment_se.txt" in the same folder correspond to the empirical target moments and their associated standard errors as reported in Table A.5 in the paper.

- This step runs all experiments using the estimated model and saves the output. Using Visual Studio, open the "fortran\benchmark\Calibration.sln" file. Set calind = 0 in the "fortran\benchmark\mod_parameters_v12.f90" file (line 10) and run_experiments = 1 and recordsimstat = 1 (lines 15 and 16). All necessary model results will be automatically saved. The total expected time is 20 minutes.
- Using Visual Studio, open the "fortran\exogenous\Calibration.sln" file and run the program once. It will generate the necessary model results for Figure 9. This takes less than 30 seconds.
- Using Matlab, open and run the "mfiles\get_figures.m" file to generate all remaining figures in the manuscript and online appendix.
- Open and run the "mfiles\get_tables.m" file to generate all remaining tables in the manuscript and online appendix.

## 5. List of Figures and Tables

The following table summarizes the figures and tables (and their location in the working paper version of the manuscript) along with their relevant program location (the final line that produces outputs) and their output location and format.

| Item | Page (Paper) | Program | Line (Code) | Output |
|---|---|---|---|---|
| Figure 1 | 6 | mfiles\get_figures.m | 150 | figures\Figure1.eps |
| Figure 2 | 8 | mfiles\get_figures.m | 186 | figures\Figure2.eps |
| Figure 3 | 10 | mfiles\get_figures.m | 218 | figures\Figure3.eps |
| Figure 4 | 21 | mfiles\get_figures.m | 251 | figures\Figure4.eps |
| Figure 5 | 28 | mfiles\get_figures.m | 290 | figures\Figure5.eps |

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|---|---|---|---|---|
| Figure 6 | 29 | mfiles\get_figures.m | 328 | figures\Figure6.eps |
| Figure 7 | 30 | mfiles\get_figures.m | 366 | figures\Figure7.eps |
| Figure 8 | 31 | mfiles\get_figures.m | 406 | figures\Figure8.eps |
| Figure 9 | 33 | mfiles\get_figures.m | 449 | figures\Figure9.eps |
| Figure 10 | 35 | mfiles\get_figures.m | 501 | figures\Figure10.eps |
| Figure 11 | 38 | mfiles\get_figures.m | 548 | figures\Figure11.eps |
| Figure A.1 | A-4 | dofiles\get_TableA2FiguresA1A2.do | 46 | figures\FigureA1pcs.eps |
| | | dofiles\get_TableA2FiguresA1A2.do | 49 | figures\FigureA1mcs.eps |
| Figure A.2 | A-5 | dofiles\get_TableA2FiguresA1A2.do | 53 | figures\FigureA2.eps |
| Figure A.3 | A-6 | mfiles\get_figures.m | 593 | figures\FigureA3.eps |
| Figure A.4 | A-20 | mfiles\get_figures.m | 640 | figures\FigureA4.eps |
| Figure A.5 | A-21 | mfiles\get_figures.m | 694 | figures\FigureA5.eps |
| Figure A.6 | A-22 | mfiles\get_figures.m | 731 | figures\FigureA6.eps |
| Figure A.7 | A-23 | mfiles\get_figures.m | 784 | figures\FigureA7.eps |
| Figure A.8 | A-23 | mfiles\get_figures.m | 825 | figures\FigureA8.eps |
| Table 1 | 11 | dofiles\get_Table1.do | 56 | tables\Table1.tex |
| Table 2 | 27 | mfiles\get_standard_errors.do | 79 | tables\Table2.tex |
| Table 3 | 31 | mfiles\get_tables.m | 86 | tables\Table3.tex |
| Table 4 | 41 | mfiles\get_tables.m | 102 | tables\Table4.tex |
| Table A.1 | A-2 | dofiles\get_TableA1.do | 42 | tables\TableA1.tex |
| Table A.2 | A-4 | dofiles\get_TableA2FiguresA1A2.do | 39 | tables\TableA2.tex |
| Table A.3 | A-5 | dofiles\get_TableA3.do | 20 | tables\TableA3.tex |
| Table A.4 | A-7 | dofiles\get_TableA4.do | 84 | tables\TableA4.tex |
| Table A.5 | A-9 | dofiles\get_TableA5.do | 231 | tables\TableA5.tex |
| Table A.6 | A-12 | dofiles\get_TableA6-A9.do | 51 | tables\TableA6.tex |
| Table A.7 | A-13 | dofiles\get_TableA6-A9.do | 126 | tables\TableA7.tex |
| Table A.8 | A-14 | dofiles\get_TableA6-A9.do | 169 | tables\TableA8.tex |

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|---|---|---|---|---|
| Table A.9 | A-15 | dofiles\get_TableA6-A9.do | 274 | tables\TableA9.tex |
| Table A.10 | A-15 | mfiles\simul_mortality.m | 155 | tables\TableA10.tex |
| Table A.11 | A-16 | mfiles\get_tables.m | 111 | tables\TableA11.tex |
| Table A.12 | A-19 | mfiles\get_tables.m | 144 | tables\TableA12.tex |
| Table A.13 | A-20 | mfiles\get_tables.m | 165 | tables\TableA13.tex |
| Table A.14 | A-22 | dofiles\get_TableA14.do | 112 | tables\TableA14.tex |

## 6. Data Citations

Goebel, J., Grabka, M. M., Liebig, S., Kroh, M., Richter, D., Schröder, C., & Schupp, J. (2019). The German socio-economic panel (SOEP). Jahrbücher für Nationalökonomie und Statistik, 239(2), 345-360.

Socio-Economic Panel (SOEP), data for years 1984-2018, version 35, SOEP, 2019, doi:10.5684/soep.v35. https://www.diw.de/en/diw_01.c.742256.en/edition/soep-core_v35__data_1984-2018.html